
CSC 220

HASH FUNCTION

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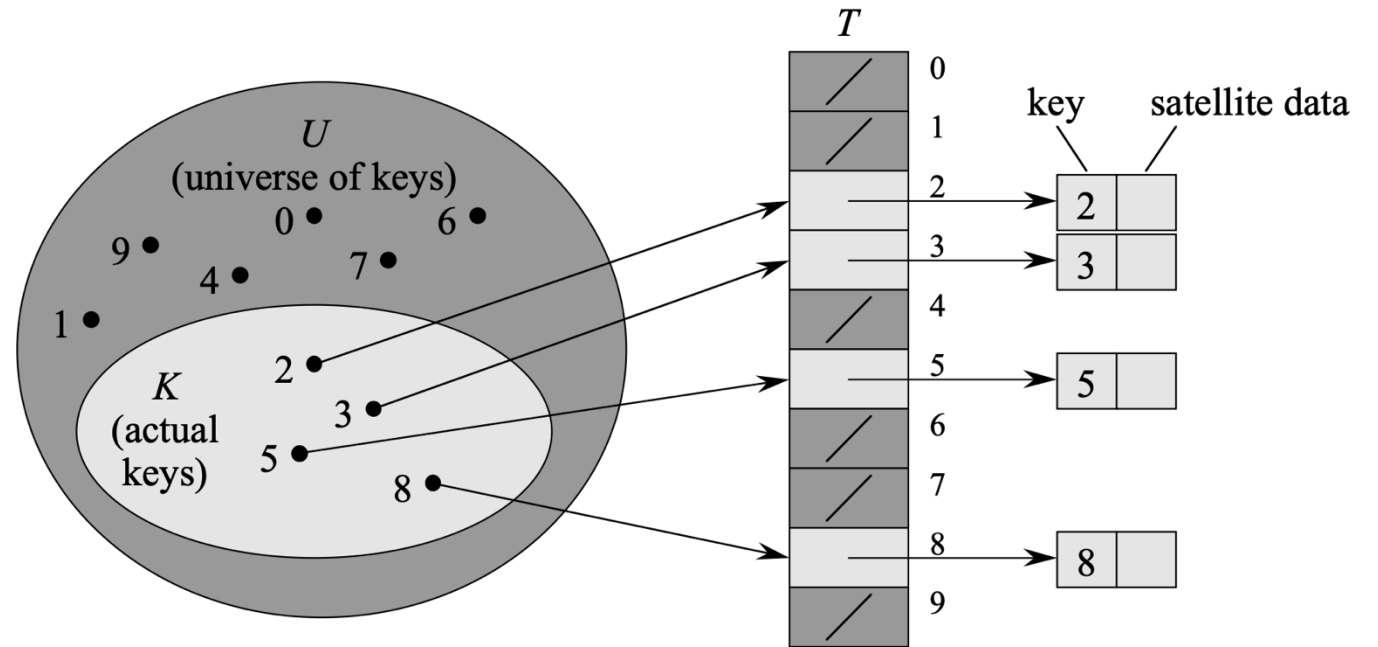
MAP: THE ABSTRACT IDEA, HASHMAP: IMPLEMENTED CLASS OF MAP

- **A map**
 - directly associates each key with a value and
 - typically lets us look up a value in **constant time**.

DIRECT ADDRESSING

A hash table generalizes the simpler notion of an ordinary array.

- Direct addressing is a simple technique that works well when the universe U of keys is reasonably small.



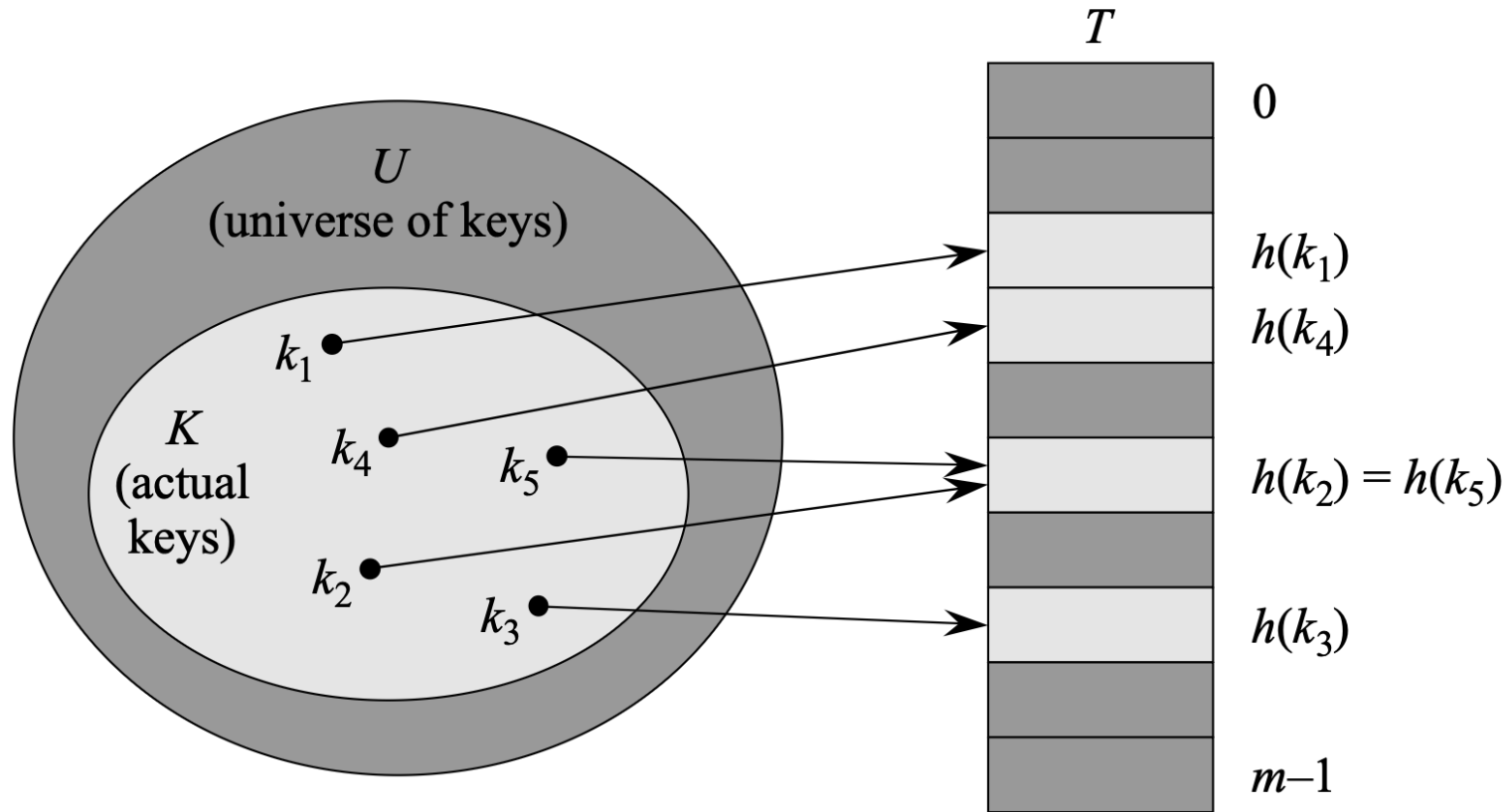
HASH TABLE

- The downside of direct addressing is obvious: if the universe U is large, storing a table T of size $|U|$ may be **impractical**, or even **impossible**, given the memory available on a typical computer.

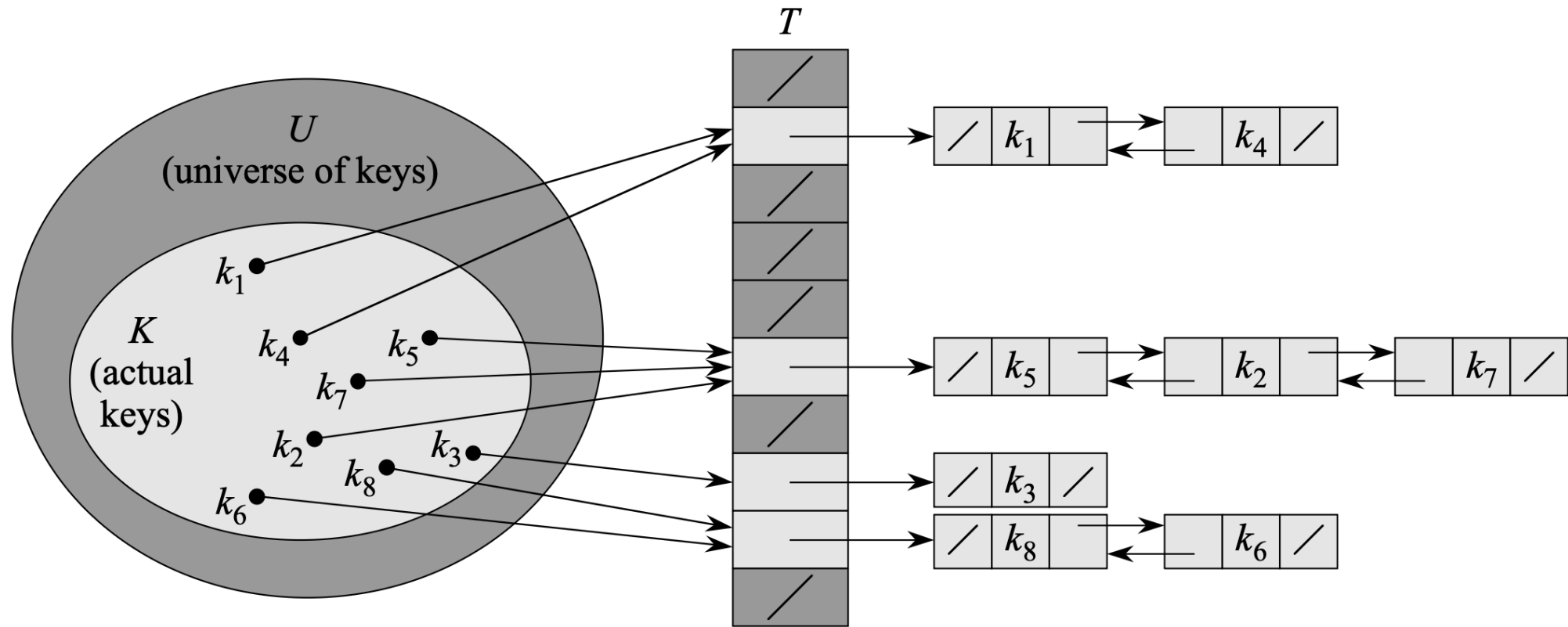
HASH TABLE AND HASH FUNCTION

- With direct addressing, an element with key k is stored in slot k .
- With hashing, this element is stored in slot $h(k)$; that is, we use a hash function h to compute the slot from the key k .

HASH TABLES & COLLISION



COLLISION RESOLUTION BY CHAINING



WHAT MAKES A GOOD HASH FUNCTION?

A good hash function satisfies (approximately) the assumption of simple **uniform hashing**:

- each key is equally likely to hash to any of the m slots,
 - independently of where any other key has hashed to.
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- Unfortunately, we typically have no way to check this condition, since we rarely know the probability distribution from which the keys are drawn. Moreover, the keys might not be drawn independently.

THE DIVISION METHOD

- In the division method for creating hash functions, we map a key k into one of m slots by taking the remainder of k divided by m .
- $h(k) = k \bmod m$

THE MULTIPLICATION METHOD

- Multiplication uses
 - $h(k) = \lfloor m (kA \bmod 1) \rfloor$, where $0 < A < 1$.
- Universal hashing randomly selects a function from a suitable family.

Method	Formula for this exercise	Final placement [m=11]
Linear	$h(k) = (k \bmod m + i) \bmod m$	22→0, 33→1, 44→2, 27→5, 38→6
Quadratic	$h(k) = (k \bmod m + i^2) \bmod m$	22→0, 33→1, 44→4, 27→5, 38→6
Double	$h(k) = (h_1(k) + ih_2(k)) \bmod m$ where $h_1(k)$ and $h_2(k)$ are two hash functions.	22→0, 33→4, 44→5, 27→2, 38→3

OPEN ADDRESSING

THE DELETION PUZZLE: $h(k) = (k \bmod 11 + i) \bmod 11$

Key	Initial index	Final index after collision resolution
22	0	0
33	0	1
44	0	2
27	5	5
38	5	6